Ups and Downs: Low-Vision Perspective on Smart Elevators and Industry Design Practices

Masaki Sawada
Tsukuba University of Technology
Tsukuba, Japan
i250105@k.tsukuba-tech.ac.jp

Yotam Sechayk
The University of Tokyo
Tokyo, Japan
sechayk-yotam@g.ecc.u-tokyo.ac.jp

Manabi Miyagi
Tsukuba University of Technology
Tsukuba, Japan
mmiyagi@k.tsukuba-tech.ac.jp

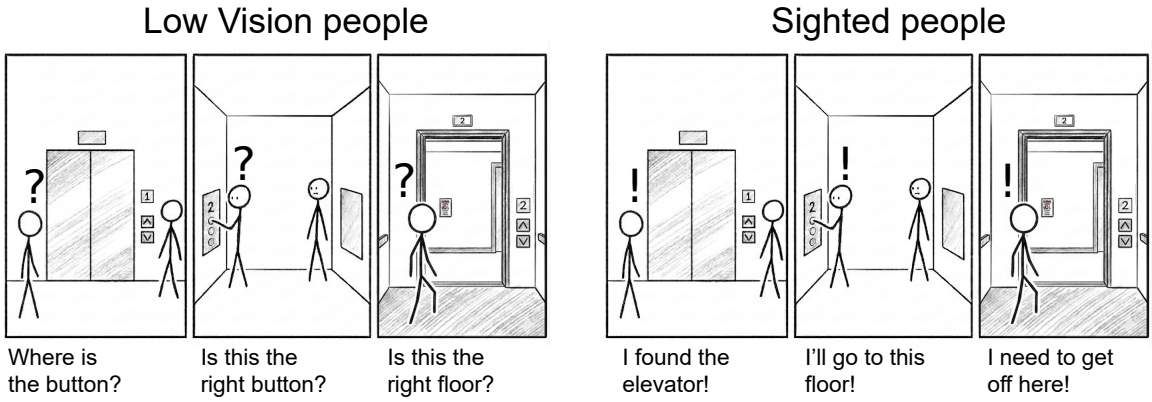


Figure 1: The challenges and difference between sighted users (right) and low-vision users (left) when using elevators.

Abstract

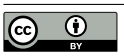
Elevators are essential for accessing workplaces, educational institutions, transportation hubs, and other public spaces. Yet the experiences of people with low vision (LV) when using elevators remain largely unexplored. In this work, we draw on the lived experience of the first author, who identifies as having LV, and reflexive conversations with three additional LV individuals and one participant whose LV progressed to blindness. We complement these lived experiences with a site visit to a major Japanese elevator manufacturer, during which the first author directly experienced and evaluated existing and emerging accessibility features. By bringing together user and industry perspectives, we reveal challenges that arise in everyday elevator use, identify gaps in existing accessibility approaches, and highlight future research and design opportunities.

CCS Concepts

• **Human-centered computing → Empirical studies in accessibility**; *Accessibility design and evaluation methods*; Accessibility technologies.

Keywords

Low vision, Elevators, Built environment accessibility, Universal design, Vertical mobility, Lived experience



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1 Introduction

Elevators are embedded in the fabric of accessible public life. From railway stations and shopping centers to office buildings and educational institutions, they provide essential vertical mobility for people with disabilities, the elderly, and anyone unable to use stairs. As public infrastructure, elevators must serve a wide range of needs, yet the gap between this ideal and real-world implementation remains significant [3, 17, 21]. In Japan, the Barrier-Free Act enacted in 2006 accelerated elevator installation in public facilities and established legal compliance standards for wheelchair users [15]. International counterparts, including EN 81-70 [1] in Europe and the ADA [24] in the United States, impose similar requirements. Events such as the 2020 Tokyo Paralympic Games and the 2025 Osaka-Kansai Expo have further raised expectations for inclusive design [7–9, 12].

Beyond physical access, elevators carry a deeper social weight. For many people with disabilities, independent elevator use is a prerequisite for equal participation in employment and public life [23]. In Japan, visually impaired individuals historically worked in occupations such as acupuncture and massage [10, 11]. However, advances in computing and assistive technology have expanded employment opportunities, making elevator accessibility an increasingly important prerequisite for workplace equity [6]. Prior work

has largely explored elevator access for users who are blind, leaving the needs of low-vision (LV) people, whose distinct challenges are not resolved by Braille buttons or voice guidance alone [20], largely unaddressed [5, 21, 29].

In this work, we explore how people with low vision experience elevator use in everyday life (Figure 1). The first author, who also identifies as having low vision, engaged in reflexive conversations with three low-vision individuals and one who transitioned from low vision to blindness. Drawing on shared lived experience, these conversations explored the challenges, adaptations, and accessibility barriers that arise when navigating various elevator systems. We complement these lived experiences with observations and hands-on interactions with accessibility technologies during a site visit to a major Japanese elevator manufacturer, enabling us to contrast everyday experiences with current industry approaches to accessibility. With these perspectives, we show that barriers span the entire elevator journey—locating the elevator, operating its controls, and verifying one's floor—and that LV users bridge them with workarounds that combine residual vision and sound. We further show that emerging technologies, such as destination dispatch systems, can remove interaction barriers while introducing new ones, and we synthesize our findings into design recommendations for elevators that better serve LV people.

2 Background

Universal design aims to produce built environments usable by all people regardless of ability; in practice, however, the built environment remains inaccessible in many ways. While the policy mandates introduced above set baseline requirements, their enforcement and interpretation remain inconsistent [5]. Audits of public buildings [4], hotel facilities [19], and metro networks [18, 27] consistently document gaps in accessibility. Even when regulations are met, they can introduce bias, such as tactile paving that excludes stairs, which limits their usage [18]. These gaps are echoed in the experience of blind and LV people: Parker et al. [16] found through interviews that major barriers in the built environment include inaccessible signage or problems finding a specific location, and Low et al. [14] found that access to information remains a significant barrier in public transport.

A complementary line of work develops navigation tools for blind and LV people. Legge et al. [13] developed a digital sign system, in which digitally-encoded signs placed throughout a building are read aloud by a handheld camera-based reader paired with a talking map; blind and LV participants used it to independently locate rooms on a single floor, though slower than sighted controls. Such wayfinding tools could help blind and LV people locate elevators, but they stop at the elevator door and do not support the interaction with the elevator itself. Zhao et al. [29] conducted a contextual inquiry with LV participants on navigating stairs, and later used visual augmentations to improve vertical mobility [28]. Their work underscores the need to explore more environmental features that affect LV mobility.

Elevators are an essential part of everyday life. For instance, the ability to live in a high-rise building depends on elevator accessibility. Tseng et al. [22] compared Braille elevator signage practices in Taiwan and the United States, identified 21 indicators that influence signage quality, and proposed an improved signage design evaluated with 32 visually impaired individuals. Trabucco et al. [21] studied deaf and hard-of-hearing users and found that 47% considered elevators unsafe, since voice-only emergency alarm systems leave them unable to communicate if trapped. Uslan [25] developed a touch and tactile destination-based controller; however, the current design requires support for blind or LV users who are not technical. Despite this body of work, the experience of LV people when using elevators remains underexplored.

The conflation of LV with blindness is not merely imprecise; it is functionally consequential, and results in solutions that do not align with the needs of LV people [20]. In this work, we center on the experience and challenges of LV people when using elevators, and explore if their needs are addressed in current industry approaches via a visit to a major elevator manufacturer in Japan.

3 Methodology

We conducted this study through two complementary approaches: reflexive conversations with LV users, and a site visit with an elevator manufacturer, which includes hands-on experience with current accessibility approaches. Both conversations and manufacturer site visit were conducted by the first author who identifies as LV. The experience and appreciation of both elevators and access needs of the first author guided this study.

3.1 Conversations with Users

We recruited four participants from Tsukuba University of Technology, a specialized institution that primarily serves students with visual or hearing impairments (Table 1). Participants were recruited via personal connections. Our participants included three LV individuals, and one blind individual who experienced a transition from LV to blindness. We conducted semi-structured conversations after providing each participant with a full explanation of the study and obtaining written informed consent. To encourage reflexive dialogue among participants with shared lived experiences, conversations were conducted in two small groups. The first group included P1 and P2, while the second included P3 and P4.

We recorded the conversations and manually transcribed them for analysis. We used reflexive thematic analysis [2] which grounds the experiences of the first author as LV as an interpretive lens. The conversations explored inconveniences when using elevators, coping strategies, and desired features. This study was approved by our institution's Ethics Review Board.

3.2 Manufacturer Site Visit and Interview

The first author conducted a site visit and an interview with a major elevator manufacturer in Japan. They inquired on the manufacturer's disability-conscious design philosophy and universal design practices in current product development. In the visit, the first author toured the production facility and a showroom, where they experienced commercialized products hands-on. The interview was recorded and transcribed manually. During the site-visit the first author took notes, which we used to reflect on the experience.

Table 1: Participant overview. A/G = age and gender. "Onset" indicates whether the visual condition was congenital or acquired. The author and P1–P3 are low-vision, and P4 is blind.

ID	A/G	Onset	Visual condition	Braille	Cane
Author	19/M	Congenital	Glaucoma, reduced acuity, night blindness	No	Yes
P1	19/M	Congenital	Retinitis Pigmentosa, partial visual field loss, night blindness	No	No
P2	19/M	Congenital	Retinitis Pigmentosa, partial visual field loss, night blindness	Yes	Yes
P3	20/M	Congenital	Optic Atrophy, Nystagmus, hypoplasia of macula	No	No
P4	19/F	Congenital	Glaucoma	Yes	Yes

4 Findings

The first author's own experience with elevators provides the primary lens through which this study is conducted. In this section we share the first author's experience and motivation for this study, and outline themes that emerged from the conversations the first author conducted with other LV and blind people. We then report on the experience of the first author visiting a major elevator manufacturer in Japan, and their hands-on experience with commercial products.

4.1 The First Author's Experience

Since childhood, the first author has held a strong interest in elevators and accumulated experience observing the specifications and design features of many models across manufacturers. The vision of the first author deteriorated around junior high-school. Their current vision condition (the ability to distinguish whether people or objects are nearby, but not to see faces or characters without close visual proximity) surfaced new challenges with elevators. Floor buttons are difficult to read; the information display panel is hard to see; and night blindness makes it difficult to locate the control panel or determine whether other people are present. When riding see-through (*i.e.,* glass-wall) elevators, the sudden change in brightness, when the doors are opening and revealing the outside, is disorienting and they "*just freeze in place.*" See Figure 2 for an illustration of the experience of the first author when using elevators. These are not edge cases; they are routine barriers that shape every elevator encounter for the first author. The first author's passion for elevators and their access motivated their exploration with other LV people and their visit to an elevator manufacturer.

4.2 Approaching and Accessing Elevators

Difficulties begin before entering the elevator. In railway stations, P2 recounted that he did not know at which branch of the tactile paving he should turn to reach the elevator, resulting in significant search time. He contrasted this with ticket gates: "*For the ticket gate you can clearly tell, since there's a beeping sound. But for the elevator there's no auditory signal or voice.*" P3 stated that at railway stations, he prefers to wait until after the train departs and the crowd thins before searching for the elevator. This is because finding the elevator itself is visually difficult in a crowd, and he wants to avoid both the risk of physical collisions and psychological stress. The first author similarly uses tactile paving as a guide toward elevators, but since it is impossible to know which branch leads to the elevator and backtracking from a wrong branch wastes time, they frequently fall back to stairs or escalators instead. For P4, beyond physical barriers, using elevators in crowded places presents psychological and social barriers. P4 mentioned, "*During crowded times I use the stairs, since elevators are tight spaces and I might bump into strangers.*"

4.3 Influence of Elevator Design

Design choices in elevator aesthetics directly impede use for LV users. See Figure 3 for an illustration of elevator design barriers. P1, who has night blindness, described difficulty locating buttons in elevators with dark designs: "*Sometimes I'm not sure where the button to go up or down is and end up hesitating.*" The first author has similar experiences and expressed similar difficulties. P3, who is LV, mentioned the poor visibility of button labels and the floor indications panel. In crowded elevators, P3 pointed out that it becomes even more difficult to see which is the current floor. The lack of audio guidance makes it impossible to know if this is the floor they requested, or someone else's which leads to embarrassment. P2 echoed a similar sentiment and mentioned that they would sometimes choose the stairs, since "*if it takes me time just to find the button, well, it just seems faster to use the stairs.*" Using the stairs, they ascend and descend themselves, which makes it easier to keep track of their location.

4.4 Impact of Specifications on Use

Differences in elevator specifications, including the presence or absence of audio announcements, Braille or raised characters, and button confirmation sounds, significantly impact usability. P2, who is right-handed and customarily uses a white cane in his right hand, prefers not to switch hands when entering elevators. As a result, he explores the left wall with his left hand. Since the main control panel in Japanese elevators is more often on the right side of the door, this creates an inconvenience and results in significantly more time to find the control panel. Moreover, since existing regulations do not consistently enforce a specific side for button placement, the task of locating the button panel becomes hard and time-consuming. P3 remarked: "*I find it quite inconsistent whether [the buttons are] installed on the right or left side of the door. So when I approach, and don't find them on this side, I realize they must be on the other. I'm not sure if this varies by manufacturer, but I think standardization would help.*"

P2, P3, and P4 all cited the lack of audio announcements and Braille labeling as significant obstacles. For multi-door elevators[1] and single-door elevators[2], they mentioned the difficulty of knowing which door will open. P1, P2, and P3 confirm the opening

[1]Elevators where the entrance or exit could be on either side.
[2]Elevators that only have one entrance that also functions as the exit.



(a) Elevator calling controls. (b) Main floor selection buttons. (c) Visual indication panel. (d) Side floor selection buttons.

Figure 2: Photos simulating the experience of the first author when using the elevator at their institution. Photos from elevators at Tsukuba University of Technology, a university for the visually and hearing impaired.

direction visually upon disembarkation; P4, who is totally blind, noted that elevators where the speaker position differentiates the opening door's side provide critical directional orientation. P4 also pointed out that the lack of audio feedback on the state of a door prevents her from discerning whether the door is open or closed.

4.5 User Strategies

Participants developed workarounds to compensate for the challenges they face. When tactile pavement is missing or difficult to access, P4 asks people nearby for directions, or follows the sound of footsteps toward the elevator. In crowded situations, where their vision is overwhelmed, P3 waits for the crowd to thin before searching. In order to identify the elevator exit location, the first author stands parallel to the elevator wall. Since the opening direction is announced from different speakers depending on which door opens, they determine which direction to face based on where the sound is louder relative to their body position.

For locating and identifying floor buttons that lack contrast, P1 described their strategy when they are alone in the elevator: pressing any nearby button first, then using the backlight shining through its number, if available, to manually discern the location of the desired floor's button. "*When the inside of the elevator is too dim to see, I just press any button, and then find my floor's button using the lit button as a reference.*" However, in social situations that can lead to discomfort, and in high-rise buildings, the number of buttons could be overwhelming and make this workaround impossible.

4.6 Desired Features

All participants consistently requested voice guidance; the first author shares this need. P1 requested that beyond embossing, all buttons would have backlighting and use high contrast visuals. For the first author, backlighting would significantly support low-light situations. P2 mentioned the need for more detailed exit guidance in both audio and Braille. For instance, in railway stations: "*Track X on the right*", or "*Track Y on the left*", along with semantic guidance to which door will open (*e.g.,* "*the door opposite to where you entered will open*"). P3 advocated for simple, large, high-contrast button designs, pointing to decorative arrows and low-visibility stylistic elements as obstacles: "*I find low-visibility designs that emphasize style particularly difficult.*" He also expressed a preference for bright, simple elevator interiors.

P4, whose vision progressed to blindness, requested both Braille and embossed tactile characters directly on buttons, and explained the distinction: "*When there are embossed characters, I can feel over the top of the button and read them even when in a hurry; since I learned how numbers are written, I can read them even though I am a Braille user.*" Direct-surface embossing allows locating a button *and* identifying the floor in a single touch, reducing operation time, since users do not need to locate both Braille and button locations.

4.7 Visit to an Elevator Manufacturer

The first author conducted a visit and an interview with a major elevator manufacturer in Japan. During the interview, they discussed the manufacturer's accessibility efforts and design practices with representatives. The visit continued with a walk through the manufacturer's showroom, where the first author had hands-on experience with their commercial offering.

Manufacturer representatives explained that, when designing elevators, they incorporate accessibility recommendations throughout the design process. For instance, when developing an elevator calling system (similar to [25]), they shared that they conducted multiple usability workshops with diverse users. This resulted in several design changes. For instance, blind participants reported difficulty and disorientation when trying to locate their designated elevator, leading to a design change that prioritized elevators near the control panel that was used. In the interview, the manufacturer highlighted that "*repeated user evaluations, rather than a single testing phase*" were necessary.

Following the interview, the first author toured the manufacturer's showroom and experienced a live demonstration of its destination dispatch system. Unlike conventional elevators, the system requires passengers to enter their destination floor on a hall operating panel before boarding. The system then assigns a specific elevator car and guides the passenger to the corresponding entrance. The first author also experienced the system's accessibility modes, including audio guidance and simplified high-contrast displays. The hall operating panel additionally provides dedicated accessibility options designed to support users with different mobility and sensory needs.

During the demonstration, the first author located and operated the hall operating panel, including its touch interface, independently. Activating the accessibility modes, however, required staff assistance: the modes are triggered by pressing a wheelchair button, and the first author found it difficult to associate a wheelchair symbol with accessibility features for LV users. Once activated, the system guided them correctly to the assigned car; beyond the

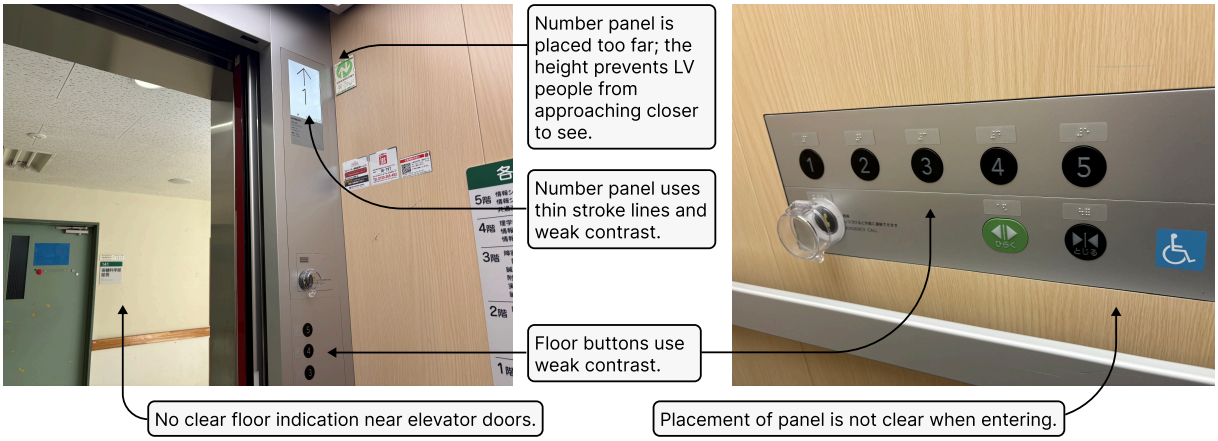


Figure 3: Interior of elevators presents several challenges to LV people: reading the floor panel, identifying location of buttons, selecting the right floor, and verifying the location upon disembark. Photos from elevators at Tsukuba University of Technology, a university for the visually and hearing impaired.

on-screen indication of the car's location, sound and lamps signaled the car's position. For the first author, specifying the destination floor at the hall removed the need to search for and press buttons inside the car, eliminating the risk of bumping into other passengers in crowded situations. At the same time, since such systems are not yet widespread, the operation method was not self-evident, and the reliance on touch panels could make operation difficult for users who cannot see the screen.

The showroom also included several smartphone-integrated elevator services. One system allowed users to call an elevator through a mobile application without interacting with a physical hall panel. This approach has the potential to eliminate one of the major barriers identified in our study: locating and operating elevator buttons. However, when the first author attempted to use the application with a screen reader, portions of the interface were not announced correctly, preventing reliable independent use. While the technology removed one accessibility barrier, it simultaneously introduced another. Representatives acknowledged this limitation and indicated plans to improve screen reader compatibility in future updates.

Reflecting on the visit, the first author observed both the opportunities and challenges of emerging elevator technologies. Features such as destination dispatch and smartphone-based elevator calling can reduce interaction burdens and improve navigation. At the same time, the accessibility of these systems depends not only on the elevator itself, but also on the accessibility of the surrounding interfaces, including mobile applications. Together with the examples discussed during the interview, these experiences reinforced the importance of evaluating accessibility throughout the entire interaction flow rather than focusing solely on the elevator hardware.

5 Designing Accessible Elevator Experiences

We explored how LV people experience elevators through reflexive conversations between the first author and four additional elevator users. Additionally, we reported on how a major elevator manufacturer in Japan incorporates accessibility into the design process, and on the hands-on experience of the first author with new developments in elevator technology. Our study surfaced several challenges that remain unaddressed, which we synthesize into design recommendations. Our recommendations highlight existing gaps that serve as opportunities for future work.

Increase access to elevator locations. One of the major challenges we identified in this work is the challenge of reaching elevators. For instance, in crowded places elevators are hard to locate, and the branching tactile pavement could lead to the stairs instead. Location of elevators should be more clearly conveyed, via sound or touch. For instance, implement opening or closing sounds for elevators or provide audio indication of tactile pavement branches as to where they lead. Precedents exist: ticket gates already emit auditory signals that P2 uses to locate them, and the Wayfindr open standard provides guidelines for audio wayfinding to lifts [26]; however, neither practice is common at elevators themselves.

Improve what is conveyed through sound. The first author and all participants expressed the need for more audio guidance. Some elevators include audio guidance; however, it is limited to declarations of direction of movement or announcement of the current floor. Audio guidance should be more descriptive, such as describing what is beyond the elevator doors, which floors have been pressed, or how many people are in the elevator. The destination dispatch system we experienced (subsection 4.7) already announces the assigned car and guides passengers to it, showing that richer announcements are commercially feasible.

Improve how sound is conveyed. The first author and most participants use the directionality of sound to orient themselves and locate places of interest. For instance, when the audio "*door is opening*," comes out of the speaker beside that door. Some multi-door elevators already differentiate the speaker position by the opening door, a cue P4 and the first author rely on; however, this

behavior is neither consistent nor standardized. More directional sound systems would aid LV people in more confidently using elevators.

Make interactive elements easier to operate. The first author and P1–P3 expressed a clear need to make interactive elements more accessible. For instance, higher contrast, larger fonts, or light shining through buttons so they can be operated even in dimly lit elevators. Partial solutions already exist: some models include backlit buttons, which P1 leverages as a workaround (subsection 4.5), and the destination dispatch system we experienced offers a simplified high-contrast display mode; however, neither is widespread. Similar to the Japanese elevator manufacturer we visited, involving LV people in the design process from the beginning and iteratively would surface these issues.

Use multiple modalities. LV users can have fluctuating or changing vision and involving multiple modalities would provide better access in such situations. For instance, P4, who experienced a decline in vision, mentioned that embossed buttons are easier than Braille, since they know how to recognize numbers. Moreover, the first author and all LV participants want to use their residual vision [20]; however, all indicate that including an audio modality would complement their vision. Standards such as EN 81-70 already mandate some redundancy, requiring raised characters and audible signals [1]; yet embossing directly on button surfaces, which P4 found fastest to use, remains uncommon. Audio, Braille, and embossing should be part of every elevator design, and other modalities should be explored.

Provide easier ways to call elevators. Our hands-on experience has revealed that work to make elevator calling more accessible is still an ongoing process. Existing approaches, such as the smartphone-based calling system we experienced (subsection 4.7), still require some manual operation and technical know-how, and their screen reader issues show that removing one barrier can introduce another. Calling elevators should be more intuitive, for instance, by automatically calling an elevator based on the proximity of users to the elevator doors.

Limitations and future work. Our participants were young adults with congenital vision conditions, recruited from a single institution. People who acquire low vision later in life, or who experience functional blindness, may develop different strategies and needs; P4's transition from LV to blindness hints at such differences. Future work should include people with acquired conditions and a wider range of visual abilities. Moreover, this study was conducted in Japan, and elevator designs, regulations, and social norms differ across countries and cultures. Cross-country and cross-cultural investigations of elevator accessibility remain an open opportunity for future work.

6 Conclusion

Elevators are essential public infrastructure, yet the experiences of people with LV remain underrepresented in accessibility research and design. Through reflexive conversations between the first author and four individuals with lived experience of visual impairment, complemented by observations of accessibility practices at a major Japanese elevator manufacturer, we examined how low-vision users navigate elevators in everyday life. Our findings reveal persistent challenges in locating elevators, interpreting available information, operating controls, and independently calling elevators. While existing accessibility features provide important support, they often fail to address the needs of users who rely on residual vision and multimodal cues. By identifying these gaps, we contribute design directions that extend elevator accessibility beyond current compliance-oriented approaches.

Future work should move beyond retrospective accounts to evaluate accessibility interventions in real-world settings and across a broader range of visual abilities and environments. Emerging technologies, including smartphone-integrated elevator systems, indoor positioning, and context-aware audio guidance, offer promising opportunities to support independent elevator use, but their accessibility and usability remain largely unexplored. More broadly, our findings highlight the importance of involving low-vision individuals throughout the design process, from early concept development to deployment and evaluation. As buildings become increasingly connected and automated, ensuring that these technologies support confident and equitable access to vertical mobility will be critical for creating truly inclusive public spaces.

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